SIWEI MENG

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SUMMARY

Third-year Computer Science Ph.D. researcher working on **Physical AI:** physical-plausibility **evaluation and benchmarking** of generative world models, **physics-based synthetic data generation**, and **physical alignment** in video generation models. Experienced in diagnostic benchmarking for physical failure discovery, multimodal preference data construction, and semantic-aware post-training of video diffusion models. Hands-on with Python/PyTorch, VLM-based physical reasoning, and confidence-guided motion probing; interested in world-model evaluation and improvement.

EDUCATION

University of Nevada, Reno, Reno, NV 2024 – Present
Ph.D. in Computer Science, GPA: 3.9/4.0, Graduate Dean's Merit Scholarship *Focus: Physical AI*

Shenzhen University, Shenzhen, China 2021 – 2024
M.E. in New Generation Electronic Information Technology, Special Freshman Scholarship *Focus: Computer Vision*

South China University of Technology 2017 – 2021
B.Eng. in Information and Interaction Design, School Second Class Scholarship *Focus: Human-Computer Interaction*

SKILLS

Programming & ML: Python, C, PyTorch, CUDA, Diffusers, HuggingFace, Linux, Git.
Evaluation & Post-Training: VLM/MLLM evaluation, DPO, GRPO, RLVR, CLIP, Video Generation/Understanding Benchmark.
Graphics and Vision: Gaussian Splatting, MPM Simulation, COLMAP, Blender, OpenCV, CUDA.
Experiment Tools: Conda, Weights & Biases, LaTeX.
Languages: Mandarin, English, Cantonese

ACADEMIC PROJECTS

PhyMetric* | *Physical Plausibility Benchmarking & Failure Diagnosis *NeurIPS ED 2026*

- Designed a scene-level diagnostic benchmark spanning **1,115 physical scenes** across mechanics, material, chemistry, thermal phenomena, fluids, and electromagnetism.
- Built an evaluation protocol with structured physical priors, question-aware frame sampling, and VLM-based binary judgments to identify specific physical failure modes beyond holistic plausibility scores.
- Evaluated multiple open-source T2V models and VLM judges, showing the importance of scene-specific QA for physical diagnosis and improved human-aligned evaluation with structured priors.

Physical & Semantic Constraint DPO* | *Video Model Post-Training *ACMMM 2026 (Oral)*

- Developed a semantic-aware DPO framework to improve physical plausibility while reducing semantic drift in text-to-video generation.
- Constructed **10K physical preference pairs** using multimodal physical evaluators, optical-flow/depth evidence, and ViCLIP-based semantic alignment scores for LoRA-based DPO training on Wan-style video models.
- Formulated constrained preference learning with per-sample semantic feasibility scores; designed the staged training strategy and semantic-aware gradient suppression to handle conflicting physical and semantic preferences

PhyMAGIC* | *Physical Reasoning & Simulation-Guided Generation *ECCV 2026*

- Developed a training-free framework for physically consistent dynamic 3D generation from a single image and text instruction.
- Designed an active motion-probing loop in which an image-to-video model generates targeted probe videos and a VLM estimates physical parameters with confidence scores.
- Integrated VLM-inferred material properties and simulator-native descriptors with **differentiable MPM simulation** initialized from 3D Gaussian reconstructions.
- Improved physical plausibility and text-motion alignment by combining generative motion evidence, VLM reasoning, and explicit physical simulation.

RESEARCH & WORK EXPERIENCE

University of Nevada, Reno Sep 2024 – Present
Graduate Research Assistant | Physical AI and Computer Vision

- Conduct research on physics-grounded generative AI across text-to-video evaluation, preference alignment, and dynamic 3D/4D content generation.
- Built diagnostic evaluation pipelines for generated videos, including scene-specific QA construction, structured physical priors, question-aware frame sampling, VLM judging, and multi-level score aggregation.

- Developed semantic-aware preference optimization methods for video diffusion models, with focus on physical plausibility, semantic fidelity, LoRA-based DPO training, and preference data construction.
- Designed training-free single-image animation pipelines that combine image-to-video motion probes, VLM-based physical reasoning, 3D Gaussian reconstruction, and MPM simulation.
- Led and coordinated collaborative research workflows, including benchmark curation, human/VLM evaluation, undergraduate annotation guidance, and edge AI research for wildfire detection.

Meitu | Research Intern Jul 2026 – Present

Long-Video Temporal Grounding, Multimodal Post-Training & Evaluation

- Developed a long-video temporal grounding benchmark covering 194 videos and 2,268 event queries, supporting fine-grained event localization and systematic evaluation of video-language models.
- Built a human–VLM annotation pipeline integrating MLLM-based event verification, temporal boundary refinement, and human review to identify and correct annotation errors.
- Developed semantic-aware preference optimization methods for video diffusion models, with focus on physical plausibility, semantic fidelity, LoRA-based DPO training, and preference data construction.
- Evaluated video-language models and analyzed temporal localization failures, investigating frame sampling and sliding-window inference to study accuracy–efficiency trade-offs in long-video understanding.

FirePrune | PSCoR HDRFS Seed Grant 2025 – 2026

Co-PI | Multi-modal Dataset and MLLM-Guided Model Pruning

- Investigated model pruning and compression strategies to reduce inference cost while preserving wildfire detection reliability on resource-constrained devices.

Parametrix.ai May – Aug 2023

AIGC Algorithm Intern

- Supported generative model training and evaluation for anime-style facial expression synthesis, with focus on complex expressions, controllable attributes, and expression consistency.
- Defined a hierarchical taxonomy of anime facial expressions to support data organization, prompt design, and fine-grained generation control.

TCL Co., Ltd. Jul – Dec 2022

Graphic Semantic Algorithm Intern

- Developed and evaluated lightweight visual recognition models for image and video understanding tasks, using YOLOv3 and MobileNet-based architectures.
- Built open/closed-eye detection models for high-speed short-video quality assessment, supporting automated screening of facial-state artifacts in social media videos.
- Conducted model testing, error analysis, and performance comparison across detection and classification settings to improve robustness under real-world visual variations.

SELECTED PUBLICATIONS

PhyMetric: Diagnosing Physical Plausibility in Text-to-Video Generation via Scene-Level QA
Siwei Meng, *Xuerou Cai, Rui Hu, Ping Liu. NeurIPS ED Track, 2026.*

PhyMAGIC: Physical Motion-Aware Generative Inference with Confidence-guided LLM
Siwei Meng, *Yawei Luo, Ping Liu. ECCV 2026.*

Physical and Semantic Constraint DPO for Text-to-Video Generation
Siwei Meng, *Yawei Luo, Shu Zhang, Ping Liu. ACMMM, 2026 (Oral).*

Grounding Creativity in Physics: A Brief Survey of Physical Priors in AIGC
Siwei Meng, *Yawei Luo, Ping Liu. IJCAI Survey Track, 2025.*

A Comprehensive Survey of Interaction Techniques in 3D Scene Generation
Yuqi Li, ***Siwei Meng,*** *Chuanguang Yang, et al. IJCAI 2026 Survey Track.*

Occlusion-aware Learning and Facial Action Units Prior for Facial Expression Recognition
Siwei Meng, *Wuzhen Shi, Yang Wen. ICME, 2026.*

Fusing Structure and Appearance Features in Facial Expression Recognition Transformer
Siwei Meng, *Wuzhen Shi. ICASSP, 2024.*